
CS246: Database Management Systems Lab

Week # 01 (1 Questions, 100 Marks)

Lab session: AL1

Held on: 06-Jan-2025 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 2

Submission time: 16:45 Hrs, 06-Jan-2025

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Question 1: (100 points)

Lab theme: **Sorting**. Write a C program to perform the distance computation of all pair of places. For each place, sort the places from closest place to farthest.

Following is the **Input file** description:

```
531022,14.0764,75.2132,A Koduru,531022,Andhra Pradesh food,Visakhapatnam
531022,14.0764,75.2132,A Kothapalle,531022,Andhra Pradesh,Visakhapatnam
530003,17.7246089,83.3185106,A U engg college,530003,Andhra Pradesh,Visakhapatnam
531027,17.895319,82.7913528,A Veerananarayanam,531027,Andhra Pradesh,Visakhapatnam
531115,17.5867769,82.4946892,Adakula,531115,Andhra Pradesh,Visakhapatnam
```

Input file `week01-input-01.txt` consists of the following fields

1. field 1: pincode
2. field 2: latitude in degrees
3. field 3: longitude in degrees
4. field 4: place
5. field 5: pincode
6. field 6: state
7. field 7: city

Your task is to compute:

Task 01 (7 marks) Declare a structure which hold the records described above whose members and data types are given below

1. pincode01: `char(6)`
2. latitude in degrees: `float`
3. longitude in degrees: `float`
4. place: `char(50)`
5. pincode02: `char(6)`

6. address: char(50)

7. city: char(20)

Task 02 (10 marks) Read the data from the input file `week01-input-01.txt`

Task 03 (23 marks) For each place p , perform the following:

1. Compute pair-wise distance between p and all places q using (latitude, longitude) information. That is compute d from p to $q = 1, 2, \dots, \#places$. The formula for computing distance in KM given (latitude, longitude) is:

$$d = 2 \times 6371 \times \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right)$$

where

(a) d = distance

(b) ϕ_1, ϕ_2 = latitude of two points in radians

(c) λ_1, λ_2 = longitude of two points in radians

(d) $\Delta\phi = \phi_2 - \phi_1$

(e) $\Delta\lambda = \lambda_2 - \lambda_1$

(f) radians = degrees $\times \frac{\pi}{180}$

Task 04 (50 marks) For each place, sort the distances from itself to all the other places in increasing order of the distance For example, A Koduru: A Kothapalle, Koruvada, ...

Task 05 (10 marks) Output format:

A Koduru: place1 (d_11 km), place2 (d_12 km), place3 (d_13 km), ...

A Kothapalle: place1 (d_21 km), place2 (d_22 km), place3 (d_23 km), ...

Koruvada: place1 (d_31 km), place2 (d_32 km), place3 (d_33 km), ...

...

where d_{11} is the distance between A Koduru and closest place1

d_{12} is the distance between A Koduru and 2nd closest place2

d_{13} is the distance between A Koduru and 3rd closest place3

...